



Linear stability of the non-zero equilibrium state for the viscous Burgers equation with time delay

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Abstract

In this article, we consider the linear stability of the non-zero equilibrium state for the viscous Burgers equation with a delay effect. The linear stability is analyzed by using the characteristic equation of the corresponding eigenvalue problem. If our equation does not have a delay effect, the characteristic equation is given by a polynomial equation. On the other hand, if our equation has a delay effect, the characteristic equation becomes a transcendental equation, and it is difficult to analyze it. In this situation, we apply the useful known result concerned with the characteristic equation for the ordinary delay differential equations and try to get the sharp stability condition for the viscous Burgers equation with delay. Furthermore, we apply the obtained results to a model of the traffic flow.

Keywords Linear stability · Viscous Burgers equation · Time delay

Mathematics Subject Classification 35B40 · 35L65 · 35E15

1 Introduction

We consider the viscous Burgers equation with a time delay

$$\partial_t \rho - \nu \partial_x^2 \rho + \partial_x (\rho V(\rho_\tau)) = 0 \quad (1.1)$$

in the one-dimensional whole space. Here $\rho = \rho(t, x)$ is an unknown function, ν is a positive constant, and $V = V(\rho)$ denotes a given smooth function that depends on ρ . In this article, we use a notation that $\rho_\tau(t, x) = \rho(t - \tau, x)$ with a positive constant τ called delay parameter.

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Letting τ tends to zero formally, (1.1) is reduced to the viscous Burgers equation

$$\partial_t \rho - \nu \partial_x^2 \rho + \partial_x (\rho V(\rho)) = 0. \quad (1.2)$$

The viscous Burgers equation is one of the important physical models in fluid mechanics, and we had already known a lot of results for (1.2) (for example, see [7, 17] etc.). On the other hand, the viscous Burgers equation with time delay is a relatively new research topic, and there are few known results. Furthermore, there are various expressions of the Burgers equations with time delay (see [6, 8, 16] etc.), and we focus on the viscous Burgers equation whose nonlinear term has a time delay in this article.

The initial-boundary value problem of the viscous Burgers equation with delay has been studied well. Indeed, Liu [11], and Tang and Wang [15] considered the stability of the problem with zero-Dirichlet boundary condition, and Tang [14] studied the non-local delay effect. Compared with the result for the problem of the bounded domain, there are few results for the one of the unbounded domain. Recently, Kubo and Ueda [10] treated the Cauchy problem of (1.1) and obtained the nonlinear stability of the zero solutions. Furthermore, they derived the condition for the initial history and the time delay to get the nonlinear stability. We remark that their result does not assign the smallness for both the initial history and the time delay. On the other hand, there is no result for the stability of the non-zero equilibrium state for (1.1), and our main purpose is to investigate the condition to get the linear stability of the non-zero equilibrium state.

This article also discusses the stability analysis of traffic flow as an application of the main result for (1.1) in Sect. 4. For traffic flow, discrete models are well analyzed. We refer the reader to [2, 3, 9] for example. The discrete model is described as the system of ordinary differential equations. In general, the discrete model is appropriate for the microscopic situations, and the continuous model is appropriate for the macroscopic situations. Nowadays, some results are discussing the rigorous derivation of the macroscopic model from the microscopic model, such as [1, 4].

To derive the linear stability for the nonlinear equation, we analyze the corresponding eigenvalue problem of the linearized equation. Let $\rho_* \in \mathbb{R}$ be an equilibrium state, and we introduce a new function $u := \rho - \rho_*$. Then the Eq. (1.1) is rewritten as

$$\partial_t u - \nu \partial_x^2 u + V(\rho_* + u_\tau) \partial_x u + (\rho_* + u) V'(\rho_* + u_\tau) \partial_x u_\tau = 0. \quad (1.3)$$

Therefore the linearized equation of (1.3) is given by

$$\partial_t u - \nu \partial_x^2 u + V(\rho_*) \partial_x u + \rho_* V'(\rho_*) \partial_x u_\tau = 0. \quad (1.4)$$

To derive the corresponding eigenvalue problem, we apply the Fourier transform to (1.4) and get

$$\partial_t \hat{u} + \nu \xi^2 \hat{u} + V(\rho_*) i \xi \hat{u} + \rho_* V'(\rho_*) i \xi \hat{u}_\tau = 0, \quad (1.5)$$

where $\xi \in \mathbb{R}$ denotes Fourier variable. The Eq. (1.5) is regarded as an ordinary differential equation with time delay, and it is known that the decay property of solutions

to (1.5) is characterized by the corresponding eigenvalue problem and the following characteristic equation (for detail, see Sect. 2.)

$$\lambda + v\xi^2 + V(\rho_*)i\xi + \rho_*V'(\rho_*)i\xi e^{-\tau\lambda} = 0. \quad (1.6)$$

Note that the equation (1.6) is a transcendental equation with infinite roots. For this characteristic equation, our main result is stated as follows.

Theorem 1.1 *All the roots of (1.6) satisfy*

$$\limsup_{|\xi| \rightarrow 0} \operatorname{Re}\lambda(\xi) \leq 0, \quad \lim_{|\xi| \rightarrow \infty} \operatorname{Re}\lambda(\xi) = -\infty. \quad (1.7)$$

Furthermore, they have negative real parts for any $\xi \in \mathbb{R} \setminus \{0\}$ if and only if v , τ , ρ_* and V satisfy one of the following three conditions:

(i)

$$\frac{\tau}{v}\rho_*V'(\rho_*)(V(\rho_*) + \rho_*V'(\rho_*)) \leq 1, \quad \frac{\tau}{v}\rho_*V(\rho_*)V'(\rho_*) \geq -\frac{\pi}{2}. \quad (1.8)$$

(ii)

$$\frac{\tau}{v}\rho_*V'(\rho_*)(V(\rho_*) + \rho_*V'(\rho_*)) \leq 1, \quad -\frac{3\pi}{2} < \frac{\tau}{v}\rho_*V(\rho_*)V'(\rho_*) < -\frac{\pi}{2}, \quad (1.9)$$

and

$$\frac{\tau}{v}V(\rho_*)^2 \left(\sqrt{\frac{\tau^2}{v^2}\rho_*^2V(\rho_*)^2V'(\rho_*)^2 - \frac{\pi^2}{4}} + \frac{\tau}{v}\rho_*^2V'(\rho_*)^2 - 1 \right) \leq \frac{\pi^2}{2}. \quad (1.10)$$

(iii) *Inequalities (1.9),*

$$\frac{\tau}{v}V(\rho_*)^2 \left(\sqrt{\frac{\tau^2}{v^2}\rho_*^2V(\rho_*)^2V'(\rho_*)^2 - \frac{\pi^2}{4}} + \frac{\tau}{v}\rho_*^2V'(\rho_*)^2 - 1 \right) > \frac{\pi^2}{2}, \quad (1.11)$$

and

$$\frac{\tau}{v}\rho_*V'(\rho_*)(\rho_*V'(\rho_*)\theta_* - V(\rho_*))\sqrt{1 - \theta_*^2} < \frac{3\pi}{2} - \operatorname{Arccos}(\sqrt{1 - \theta_*^2}). \quad (1.12)$$

Here, θ_* is defined by

$$\theta_* := \frac{1}{4|\rho_* V'(\rho_*)|} \left(\sqrt{V(\rho_*)^2 + 8 \left(\rho_*^2 V'(\rho_*)^2 + \frac{\nu}{\tau} \right)} - |V(\rho_*)| \right).$$

Theorem 1.1 tells us that the delay effect induces the instability phenomena. For example, if the delay parameter τ satisfies $\tau > -\nu\pi/(2\rho_* V(\rho_*) V'(\rho_*))$ if $\rho_* V(\rho_*) V'(\rho_*) < 0$, there exists $\xi_0 \in \mathbb{R}$ such that $\operatorname{Re}\lambda(\xi_0) \geq 0$, and the equilibrium state ρ_* will be unstable.

This article is organized as follows. We will show a key lemma and give the proof of the main results in Sect. 2. In Sect. 3, we will remark about the conditions mentioned in Theorem 1.1. As an application of Theorem 1.1, we introduce a traffic flow model and discuss the stability condition in Sect. 4.

2 Proof of the main results

Our purpose is to derive the property of the eigenvalue λ of (1.6). To this end, we employ the theorem derived by Nishiguchi [12] for

$$\lambda + \alpha - \beta e^{-\tau\lambda} = 0. \quad (2.1)$$

Theorem 2.1 [12] *Let $\alpha \in \mathbb{C}$, $\beta \in \mathbb{C} \setminus \{0\}$, and $\tau > 0$. Then all the roots of (2.1) have negative real parts if and only if α , β , and τ satisfy the following (i) or (ii):*

- (i) $\operatorname{Re}(\alpha) > |\beta|$.
- (ii) $-|\beta| < \operatorname{Re}(\alpha) \leq |\beta|$ and

$$Z(\operatorname{Im}(\alpha)\tau + \operatorname{Arg}(\beta)) > \operatorname{Arccos}(\operatorname{Re}(\alpha)/|\beta|) + \tau\sqrt{|\beta|^2 - \operatorname{Re}(\alpha)^2}.$$

Here, $Z(x) := \operatorname{Arccos}(\cos(x))$.

We remark that Arccos denotes the inverse trigonometric function of cosine, and $Z(x)$ satisfies

$$Z(x) = \begin{cases} -x + 2k\pi, & x \in [(2k-1)\pi, 2k\pi], \\ x - 2k\pi, & x \in [2k\pi, (2k+1)\pi] \end{cases}$$

for $k \in \mathbb{Z}$. From Theorem 2.1, we derive the following lemma.

Lemma 2.2 *Let $a \in \mathbb{R}$, $b > 0$, and define*

$$f(y) := Z(ay - \pi/2), \quad g(y) := \operatorname{Arccos}(y) + by\sqrt{1-y^2} \quad (2.2)$$

for $y \in [0, 1]$. Then, the following statements hold.

- (i) If $a + b - 1 > 0$ or $a \leq -3\pi/2$, then there exists a non-empty set $I \subset (0, 1]$ such that $f(y) < g(y)$ for $y \in I$.
- (ii) If $a + b - 1 \leq 0$ and $a \geq -\pi/2$, then $f(y) > g(y)$ for $y \in (0, 1]$.
- (iii) If $a + b - 1 \leq 0$, $-3\pi/2 < a < -\pi/2$, and

$$-\pi^2 b/(2a^2) - a\sqrt{1 - \pi^2/(4a^2)} + b - 1 \leq 0, \quad (2.3)$$

then $f(y) > g(y)$ for $y \in (0, 1]$.

- (iv) If $a + b - 1 \leq 0$, $-3\pi/2 < a < -\pi/2$, and

$$-\pi^2 b/(2a^2) - a\sqrt{1 - \pi^2/(4a^2)} + b - 1 > 0, \quad (2.4)$$

then the following cases occur.

- (iv-i) If $f(y_*) > g(y_*)$, then $f(y) > g(y)$ for $y \in (0, 1]$.
- (iv-ii) If $f(y_*) = g(y_*)$, then $f(y) > g(y)$ for $y \in (0, 1] \setminus \{y_*\}$.
- (iv-iii) If $f(y_*) < g(y_*)$, then there exists a non-empty set $J \subset (0, 1]$ such that $f(y) < g(y)$ for $y \in J$.

Here, $y_* \in (0, 1]$ is defined by

$$y_* := \sqrt{1 - \frac{1}{16b^2} \left(a + \sqrt{a^2 + 8b(b+1)} \right)^2}. \quad (2.5)$$

Lemma 2.3 *If the functions $f(y)$ and $g(y)$ defined by (2.2) satisfy $f(y) > g(y)$ for $y \in (0, 1]$, then the pair of $a \in \mathbb{R}$ and $b > 0$ satisfies one of the following (i) to (iii).*

- (i) It holds that $a + b - 1 \leq 0$ and $a \geq -\pi/2$.
- (ii) It holds that $a + b - 1 \leq 0$, $-3\pi/2 < a < -\pi/2$, and (2.3).
- (iii) It holds that $a + b - 1 \leq 0$, $-3\pi/2 < a < -\pi/2$, (2.4), and $f(y_*) > g(y_*)$, where y_* is defined by (2.5).

Since Lemmas 2.2 and 2.3, we obtain the necessary and sufficient conditions related to f , g and the pair of a , b . Lemma 2.3 is proved by the same analogy as the proof of Lemma 2.2, and we omit the detail.

We introduce the sets

- (i) $\mathcal{U} := \{(a, b) \in \mathbb{R} \times (0, \infty) \mid a + b - 1 > 0\}$
 $\cup \{(a, b) \in \mathbb{R} \times (0, \infty) \mid a \leq -3\pi/2\},$
- (ii) $\mathcal{S}_1 := \{(a, b) \in \mathbb{R} \times (0, \infty) \mid a + b - 1 \leq 0, a \geq -\pi/2\}, \quad (2.6)$
- (iii) $\mathcal{S}_2 := \{(a, b) \in \mathbb{R} \times (0, \infty) \mid a + b - 1 \leq 0, -3\pi/2 < a < -\pi/2, h(a, b) \leq 0\},$
- (iv) $\mathcal{T} := \{(a, b) \in \mathbb{R} \times (0, \infty) \mid a + b - 1 \leq 0, -3\pi/2 < a < -\pi/2, h(a, b) > 0\},$

and $\mathcal{S} := \mathcal{S}_1 \cup \mathcal{S}_2$, where $h(a, b) := -\pi^2 b/(2a^2) - a\sqrt{1 - \pi^2/(4a^2)} + b - 1$. These sets correspond to each of the statements in Lemma 2.2. Then, noting that (2.3) is

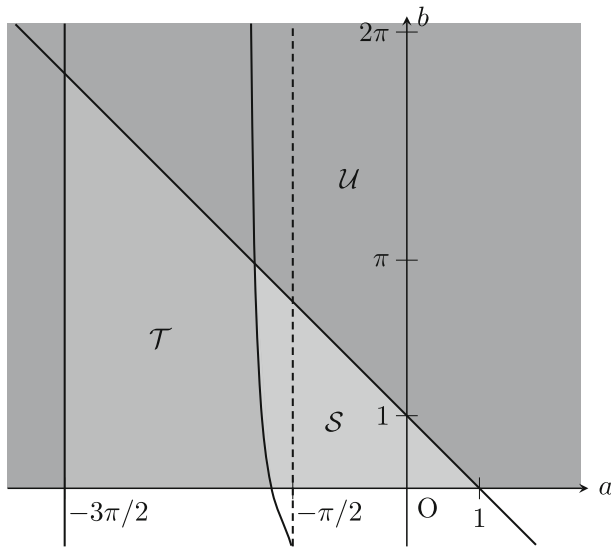


Fig. 1 The subsets $\mathcal{U}$, $\mathcal{S}$, and $\mathcal{T}$ given in (2.6)

rewritten as

$$b \geq \frac{a^2(2 - \sqrt{4a^2 - \pi^2})}{2a^2 - \pi^2} \quad \text{if } a^2 < \pi^2/2,$$

$$b \leq \frac{a^2(2 - \sqrt{4a^2 - \pi^2})}{2a^2 - \pi^2} < 0 \quad \text{if } a^2 > \pi^2/2$$

for $-3\pi/2 < a < -\pi/2$, the subsets $\mathcal{U}$, $\mathcal{S}$, and $\mathcal{T}$ are described as in Fig. 1.

Proof of Lemma 2.2 The function $f(y)$ satisfies $f(0) = \pi/2$ and $f'(0) = -a$. Furthermore, in the case of $a \neq 0$, the definition of $f(y)$ gives

$$f(y) = \begin{cases} -ay + (2k + 1/2)\pi, & ay \in [(2k - 1/2)\pi, (2k + 1/2)\pi], \\ ay - (2k + 1/2)\pi, & ay \in [(2k + 1/2)\pi, (2k + 3/2)\pi] \end{cases}$$

for $k \in \mathbb{Z}$, and hence

$$f((2k + 1/2)\pi/a) = 0, \quad f((2k - 1/2)\pi/a) = \pi.$$

On the other hand, the function $g(y)$ satisfies $g(0) = \pi/2$, $g(1) = 0$, and

$$g'(y) = \frac{b - 1 - 2by^2}{\sqrt{1 - y^2}}, \quad g''(y) = \frac{y(2by^2 - 3b - 1)}{(1 - y^2)\sqrt{1 - y^2}}.$$

Thus we have $g'(0) = b - 1$ and $\lim_{y \rightarrow 1-0} g'(y) = -\infty$. Furthermore, we also find that $g''(0) = 0$ and $g''(y) < 0$ for $y \in (0, 1)$. Namely, $g(y)$ is concave function on $[0, 1]$.

Firstly, we consider the case $a + b - 1 > 0$. In this case, we obtain $f'(0) < g'(0)$. Therefore, because of $f(0) = g(0)$, there exists the interval $I_1 \subset (0, 1]$ such that $f(y) < g(y)$ for $y \in I_1$ (see, for example, Fig. 2 for the case of $a = 0.5$, $b = 1$).

Secondly, we consider the case $a \leq -3\pi/2$. In this case, since $f(-3\pi/(2a)) = 0$ and $g(1) = 0$, there exists the interval $I_2 \subset (0, 1]$ such that $f(y) < g(y)$ for $y \in I_2$.

Thirdly, we consider the case $a + b - 1 \leq 0$. Now, we remark that $f'(0) \geq g'(0)$. If $a \geq -\pi/2$, then since $a + b - 1 \leq 0$ and $b > 0$ imply $-\pi/2 \leq a \leq 1$, we have $f(y) = -ay + \pi/2$ for $y \in [0, 1]$. Therefore, $f(y) > g(y)$ holds for $y \in (0, 1]$ (see, for example, Figs. 3 for the case of $a = 0.5$, $b = 0.3$). If $-3\pi/2 < a < -\pi/2$, we note that $f(y) > g(y)$ is satisfied for $y \in (0, -\pi/(2a)]$ by the same argument as before. Thus, we focus on the tangent point of $f(y)$ and $g(y)$ for $y \in (-\pi/(2a), 1)$. The tangent point must satisfy

$$a = \frac{b - 1 - 2by^2}{\sqrt{1 - y^2}}. \quad (2.7)$$

The new variable $z = \sqrt{1 - y^2}$ leads that (2.7) is equivalent to $\tilde{h}(z) = 0$, where $\tilde{h}(z) := 2bz^2 - az - b - 1$. We find that

$$\tilde{h}(0) = -b - 1, \quad \tilde{h}(\sqrt{1 - \pi^2/(4a^2)}) = -\pi^2 b/(2a^2) - a\sqrt{1 - \pi^2/(4a^2)} + b - 1.$$

Thus, there is a root of $\tilde{h}(z) = 0$ for $z \in (\sqrt{1 - \pi^2/(4a^2)}, 1)$ if

$$-\pi^2 b/(2a^2) - a\sqrt{1 - \pi^2/(4a^2)} + b - 1 > 0, \quad (2.8)$$

and this root is described as

$$z_* = \frac{1}{4b}(a + \sqrt{a^2 + 8b(b + 1)}). \quad (2.9)$$

Consequently, we obtain the following conclusion. Since $f(y)$ is linear and $g(y)$ is concave, $f(y) > g(y)$ is satisfied for $y \in (-\pi/(2a), 1)$ if

$$-\pi^2 b/(2a^2) - a\sqrt{1 - \pi^2/(4a^2)} + b - 1 \leq 0.$$

On the other hand, in the case (2.8), we have three situations that

$$\begin{aligned} f(y) &> g(y) & \text{for } y \in (-\pi/(2a), 1) & & \text{if } f(y_*) > g(y_*), \\ f(y) &> g(y) & \text{for } y \in (-\pi/(2a), 1) \setminus \{y_*\} & & \text{if } f(y_*) = g(y_*), \\ f(y) &< g(y) & \text{for } y \in \exists J \subset (-\pi/(2a), 1) & & \text{if } f(y_*) < g(y_*), \end{aligned}$$

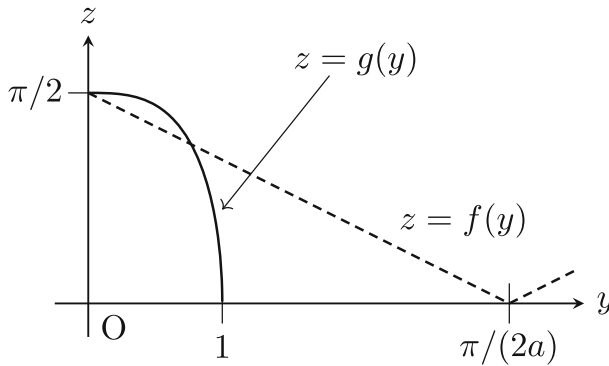


Fig. 2 $a = 0.5, b = 1$

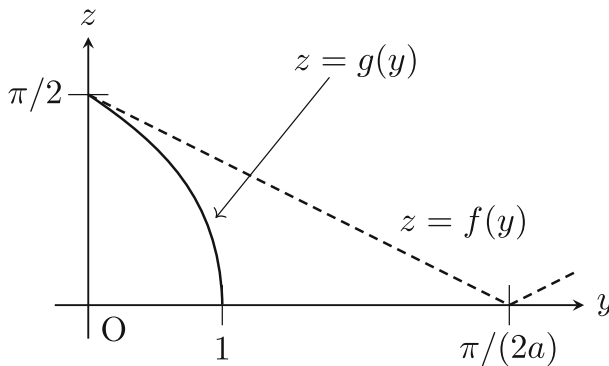


Fig. 3 $a = 0.5, b = 0.3$

where $y_* = \sqrt{1 - z_*^2}$. Therefore, we conclude the statement (iv). See Figs. 4 and 5 for help in understanding the situation.

We shall give a small remark on Lemma 2.2. Because of $y_* \in (-\pi/(2a), 1)$, and $f(y) = ay + 3\pi/2$ for $y \in [-\pi/(2a), -3\pi/(2a)]$, the condition $f(y_*) > g(y_*)$ is reduced to

$$ay_* + \frac{3\pi}{2} > \operatorname{Arccos}(y_*) + by_*\sqrt{1 - y_*^2}. \quad (2.10)$$

This representation is used in the proof of Theorem 1.1.

Proof of Theorem 1.1 By Rouché's theorem, we find that the roots λ of (1.6) are continuous for τ and ξ . For the detail, we refer the reader to [13]. The real and imaginary parts of eigenvalues of (1.6) satisfy

$$\operatorname{Re} \lambda + \nu \xi^2 + \rho_* V'(\rho_*) \xi e^{-\tau \operatorname{Re} \lambda} \sin(\tau \operatorname{Im} \lambda) = 0,$$

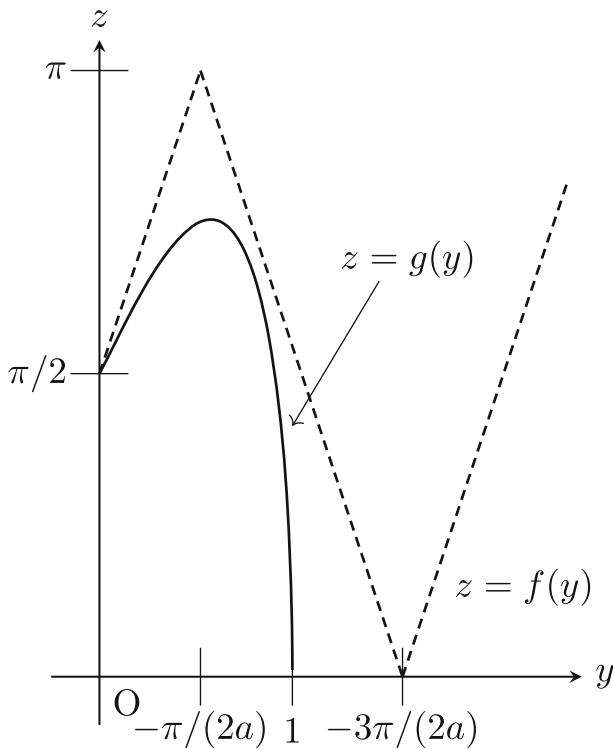


Fig. 4 $a = -3, b = 3$

and this gives

$$\operatorname{Re} \lambda + \nu \xi^2 \leq |\rho_* V'(\rho_*)| |\xi| e^{-\tau \operatorname{Re} \lambda}. \quad (2.11)$$

To prove (1.7), we consider the negation of the conditions in (1.7). The negation of the first estimate in (1.7) is as follows.

$$\exists \varepsilon_1 > 0, \quad \forall \delta > 0, \quad \exists \xi_1 \in [-\delta, \delta] \text{ s.t. } \operatorname{Re} \lambda(\xi_1) \geq \varepsilon_1.$$

Let $\delta = \varepsilon_1 / (2|\rho_* V'(\rho_*)|)$. Then (2.11) leads to

$$\operatorname{Re} \lambda(\xi_1) + \nu \xi_1^2 \leq |\rho_* V'(\rho_*)| |\xi_1| e^{-\tau \operatorname{Re} \lambda(\xi_1)} \leq |\rho_* V'(\rho_*)| \delta = \frac{\varepsilon_1}{2},$$

and this is a contradiction. Similarly, the negation of the second condition in (1.7) is as follows.

$$\exists \varepsilon_2 > 0, \quad \forall \delta > 0, \quad \exists \xi_2 \in (-\infty, -\delta] \cup [\delta, \infty) \text{ s.t. } \operatorname{Re} \lambda(\xi_2) > -\varepsilon_2.$$

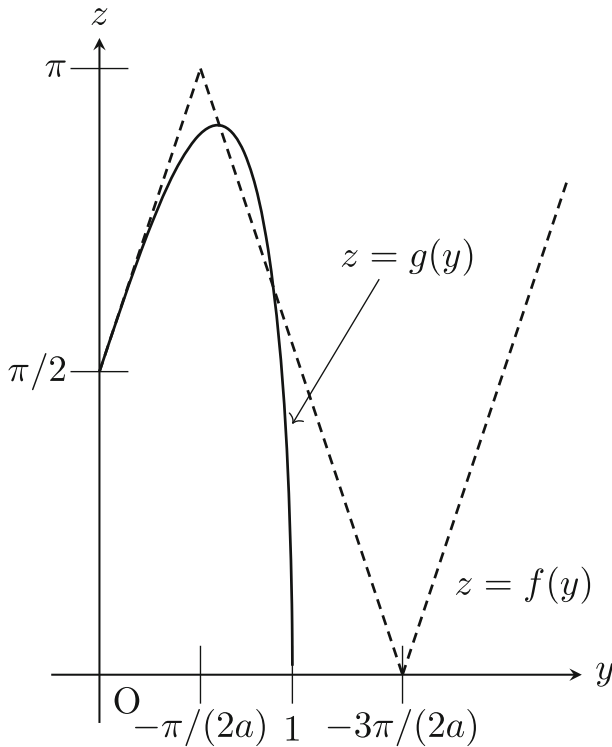


Fig. 5 $a = -3, b = 4$

Let $\delta = \max\{2, (\varepsilon_2 + |\rho_* V'(\rho_*)|e^{\tau\varepsilon_2})/\nu\}$. Then (2.11) leads to

$$\begin{aligned} \operatorname{Re}\lambda(\xi_2) &\leq -\nu\xi_2^2 + |\rho_* V'(\rho_*)||\xi_2|e^{-\tau\operatorname{Re}\lambda(\xi_2)} \leq -|\xi_2|(\nu\delta - |\rho_* V'(\rho_*)|e^{\tau\varepsilon_2}) \\ &\leq -\delta\varepsilon_2 \leq -2\varepsilon_2, \end{aligned}$$

and this is a also contradiction.

We find that $\operatorname{Re}\lambda(\xi) = -\nu\xi^2$ if $\rho_* V'(\rho_*)\xi = 0$, and it is enough to consider the case that $\rho_* V'(\rho_*)\xi \neq 0$. We apply Theorem 2.1 to (1.6) by letting $\alpha = \nu\xi^2 + V(\rho_*)i\xi$ and $\beta = -\rho_* V'(\rho_*)i\xi$. The condition (i) in Theorem 2.1 becomes

$$|\xi| > \frac{|\rho_* V'(\rho_*)|}{\nu}.$$

The condition (ii) in Theorem 2.1 becomes

$$|\xi| \leq \frac{|\rho_* V'(\rho_*)|}{\nu} \quad (2.12)$$

and

$$\begin{aligned} Z(\tau V(\rho_*)\xi - \pi/2) &> \operatorname{Arccos}\left(\frac{\nu|\xi|}{|\rho_* V'(\rho_*)|}\right) \\ &+ \tau|\xi|\sqrt{|\rho_* V'(\rho_*)|^2 - \nu^2\xi^2} \quad \text{if } \rho_* V'(\rho_*)\xi > 0, \\ Z(\tau V(\rho_*)\xi + \pi/2) &> \operatorname{Arccos}\left(\frac{\nu|\xi|}{|\rho_* V'(\rho_*)|}\right) \\ &+ \tau|\xi|\sqrt{|\rho_* V'(\rho_*)|^2 - \nu^2\xi^2} \quad \text{if } \rho_* V'(\rho_*)\xi < 0. \end{aligned} \quad (2.13)$$

We introduce a new variable $\eta = \nu|\xi|/|\rho_* V'(\rho_*)|$. Then (2.12) and (2.13) are rewritten as $0 < \eta \leq 1$ and

$$Z\left(\frac{\tau}{\nu}\rho_* V(\rho_*)V'(\rho_*)\eta - \frac{\pi}{2}\right) > \operatorname{Arccos}(\eta) + \frac{\tau}{\nu}\rho_*^2 V'(\rho_*)^2 \eta \sqrt{1 - \eta^2}. \quad (2.14)$$

Here, we used the fact that $Z(x)$ is an even function.

To employ Lemma 2.2, we take

$$a = \frac{\tau}{\nu}\rho_* V(\rho_*)V'(\rho_*), \quad b = \frac{\tau}{\nu}\rho_*^2 V'(\rho_*)^2. \quad (2.15)$$

Then, by applying (ii) of Lemma 2.2, it holds that if ν , τ , ρ_* , and V satisfy (1.8), then all the roots of (1.6) have negative real parts for any ξ with (2.12). On the other hand, by applying (iii) of Lemma 2.2, it holds that if ν , τ , ρ_* , and V satisfy (1.9) and (1.10), then all the roots of (1.6) have negative real parts for any ξ with (2.12). Furthermore, (iv-i) in Lemma 2.2 gives that, if (1.9), (1.11) and

$$Z\left(\frac{\tau}{\nu}\rho_* V(\rho_*)V'(\rho_*)\eta_* - \frac{\pi}{2}\right) > \operatorname{Arccos}(\eta_*) + \frac{\tau}{\nu}\rho_*^2 V'(\rho_*)^2 \eta_* \sqrt{1 - \eta_*^2}, \quad (2.16)$$

the inequality (2.14) is also satisfied, where η_* is defined by

$$\eta_* := \sqrt{1 - \frac{1}{16\rho_*^2 V'(\rho_*)^2} \left(\sqrt{V(\rho_*)^2 + 8\left(\rho_*^2 V'(\rho_*)^2 + \frac{\nu}{\tau}\right)} - |V(\rho_*)| \right)^2}.$$

By using (2.10), the condition (2.16) is reduced to

$$\frac{\tau}{\nu}\rho_* V(\rho_*)V'(\rho_*)\eta_* + \frac{3\pi}{2} > \operatorname{Arccos}(\eta_*) + \frac{\tau}{\nu}\rho_*^2 V'(\rho_*)^2 \eta_* \sqrt{1 - \eta_*^2}.$$

Consequently, taking $\theta_* = \sqrt{1 - \eta_*^2}$, we obtain the desired conditions in Theorem 1.1 and complete the proof.

3 Stability condition

In this section, we characterize the conditions in Theorem 1.1. We define a and b as in (2.15). Then the sets $\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{T}$ in (2.6) correspond to the conditions (i), (ii) and (iii) in Theorem 1.1, respectively. Namely, $\mathcal{S}$ and $\mathcal{U}$ denote the stable and unstable regions, respectively. Furthermore, $\mathcal{T}$ is separated as the stable and unstable regions by (1.12), which comes from

$$(bz_* - a)\sqrt{1 - z_*^2} < \frac{3\pi}{2} - \text{Arccos}(\sqrt{1 - z_*^2}).$$

Here, z_* is defined by (2.9). See Fig. 1 for a detailed depiction of the situation.

4 Application to a traffic flow model

In this section, we treat a simple traffic flow model as an application of Theorem 1.1. The model of traffic flow is described by the viscous Burgers equation (1.2) with a velocity function

$$V(\rho) = V_m \left(1 - \frac{\rho}{\rho_m} \right). \quad (4.1)$$

Here, the unknown function $\rho = \rho(t, x)$ denotes the traffic density, the positive constant ν is the diffusion coefficient, and the positive constants V_m and ρ_m denote the maximum speed as $\rho \rightarrow 0$ and the maximum density, respectively (for example, see Garavello et al. [5]).

It is well-known that the viscous Burgers equation (1.2) is a strictly parabolic partial differential equation, so its solution has an infinite speed of propagation. In terms of traffic flow, this means that drivers and their vehicles are assumed to react instantly to changes in the density and its gradient. To improve this troubling feature, we assume that the velocity function V does not depend on ρ but ρ_τ . Namely, our new traffic flow model is represented as (1.1) with (4.1).

For the delayed traffic flow model (1.1) with (4.1), the equilibrium state ρ_* must satisfy $0 \leq \rho_* \leq \rho_m$ because the velocity function V must be non-negative. Then Theorem 1.1 gives the following result for the eigenvalues of the linearized problem of (1.1) with (4.1).

Corollary 4.1 *All the roots of (1.6) with (4.1) satisfy*

$$\limsup_{|\xi| \rightarrow 0} \text{Re} \lambda(\xi) \leq 0, \quad \lim_{|\xi| \rightarrow \infty} \text{Re} \lambda(\xi) = -\infty.$$

Furthermore, they have negative real parts for any $\xi \in \mathbb{R} \setminus \{0\}$ if and only if ν , τ , ρ_ , ρ_m and V_m satisfy one of the following three conditions:*

(i)

$$\frac{\tau \rho_* V_m^2}{\nu \rho_m} \left(\frac{2\rho_*}{\rho_m} - 1 \right) \leq 1, \quad \frac{\tau \rho_* V_m^2}{\nu \rho_m} \left(1 - \frac{\rho_*}{\rho_m} \right) \leq \frac{\pi}{2}.$$

(ii)

$$\frac{\tau \rho_* V_m^2}{\nu \rho_m} \left(\frac{2\rho_*}{\rho_m} - 1 \right) \leq 1, \quad \frac{\pi}{2} < \frac{\tau \rho_* V_m^2}{\nu \rho_m} \left(1 - \frac{\rho_*}{\rho_m} \right) < \frac{3\pi}{2}, \quad (4.2)$$

and

$$\frac{\tau V_m^2}{\nu} \left(1 - \frac{\rho_*}{\rho_m} \right)^2 \left(\sqrt{\frac{\tau^2 \rho_*^2 V_m^4}{\nu^2 \rho_m^2} \left(1 - \frac{\rho_*}{\rho_m} \right)^2 - \frac{\pi^2}{4}} + \frac{\tau \rho_*^2 V_m^2}{\nu \rho_m^2} - 1 \right) \leq \frac{\pi^2}{2}.$$

(iii) Inequalities (4.2),

$$\frac{\tau V_m^2}{\nu} \left(1 - \frac{\rho_*}{\rho_m} \right)^2 \left(\sqrt{\frac{\tau^2 \rho_*^2 V_m^4}{\nu^2 \rho_m^2} \left(1 - \frac{\rho_*}{\rho_m} \right)^2 - \frac{\pi^2}{4}} + \frac{\tau \rho_*^2 V_m^2}{\nu \rho_m^2} - 1 \right) > \frac{\pi^2}{2},$$

and

$$\frac{\tau \rho_* V_m^2}{\nu \rho_m} \left(1 - \frac{\rho_*}{\rho_m} (1 - \theta_*) \right) \sqrt{1 - \theta_*^2} < \frac{3\pi}{2} - \text{Arccos}(\sqrt{1 - \theta_*^2}). \quad (4.3)$$

Here, θ_* is defined by

$$\theta_* = \frac{\rho_m}{4\rho_*} \left(\sqrt{\left(1 - \frac{\rho_*}{\rho_m} \right)^2 + 8 \left(\frac{\rho_*^2}{\rho_m^2} + \frac{\nu}{\tau V_m^2} \right)} - \left(1 - \frac{\rho_*}{\rho_m} \right) \right).$$

Corollary 4.1 tells us that linear stability is obtained if the equilibrium density ρ_* or the delay parameter τ is suitably small. On the other hand, if the equilibrium density ρ_* or the delay parameter τ is large enough, instability should occur.

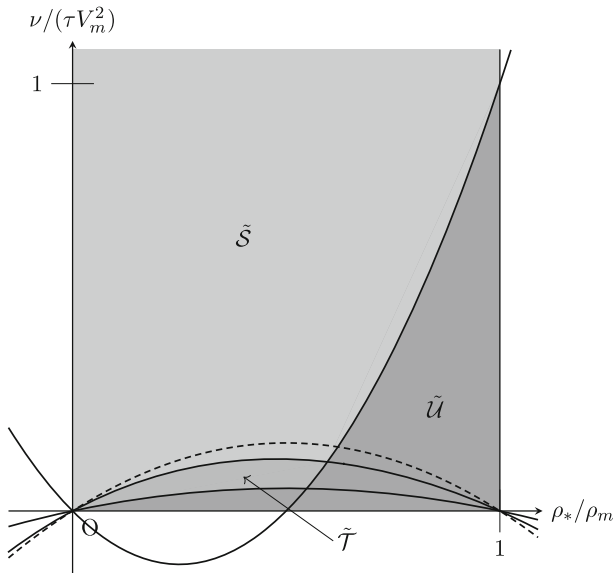


Fig. 6 Stability region in traffic flow

Furthermore, the conditions in Corollary 4.1 are characterized by the following regions.

$$\begin{aligned}\tilde{\mathcal{U}} &:= \{(\phi, \psi) \in [0, 1] \times [0, \infty) \mid \phi(2\phi - 1) > \psi\} \\ &\quad \cup \{(\phi, \psi) \in [0, 1] \times [0, \infty) \mid \phi(1 - \phi) \geq 3\pi\psi/2\}, \\ \tilde{\mathcal{S}}_1 &:= \{(\phi, \psi) \in [0, 1] \times [0, \infty) \mid \phi(2\phi - 1) \leq \psi, \phi(1 - \phi) \leq \pi\psi/2\}, \\ \tilde{\mathcal{S}}_2 &:= \{(\phi, \psi) \in [0, 1] \times [0, \infty) \mid \phi(2\phi - 1) \leq \psi, \\ &\quad \pi\psi/2 < \phi(1 - \phi) < 3\pi\psi/2, k(\phi, \psi) \leq \pi^2/2\}, \\ \tilde{\mathcal{T}} &:= \{(\phi, \psi) \in [0, 1] \times [0, \infty) \mid \phi(2\phi - 1) \leq \psi, \\ &\quad \pi\psi/2 < \phi(1 - \phi) < 3\pi\psi/2, k(\phi, \psi) > \pi^2/2\},\end{aligned}$$

and $\tilde{\mathcal{S}} := \tilde{\mathcal{S}}_1 \cup \tilde{\mathcal{S}}_2$, where $\phi := \rho_*/\rho_m$, $\psi := \nu/(\tau V_m^2)$, and

$$k(\phi, \psi) := \frac{(1 - \phi)^2}{\psi} \left(\sqrt{\frac{\phi^2}{\psi^2}(1 - \phi)^2 - \frac{\pi^2}{4} + \frac{\phi^2}{\psi} - 1} \right).$$

Here, $\tilde{\mathcal{S}}$ and $\tilde{\mathcal{U}}$ denote the stable and unstable regions, respectively. $\tilde{\mathcal{T}}$ is separated as the stable and unstable regions by the condition (4.3) (see Fig. 6).

Remark that the set constructed by condition (iii) in Corollary 4.1 is not empty. Indeed, if we take $\rho_*/\rho_m = 1/2$, then the condition (iii) gives

$$2\pi < \frac{\tau V_m^2}{\nu} < 6\pi, \quad \frac{\tau V_m^2}{\nu} \left(\sqrt{\frac{\tau^2 V_m^4}{\nu^2} - 4\pi^2} + \frac{\tau V_m^2}{\nu} - 4 \right) > 8\pi^2, \quad (4.4)$$

and

$$\frac{\tau V_m^2}{\nu} < \frac{2(3\pi - 2\text{Arccos}(\sqrt{1 - \theta_*^2}))}{(1 + \theta_*)\sqrt{1 - \theta_*^2}}, \quad \theta_* = \frac{1}{2} \left(\sqrt{\frac{9}{4} + \frac{8\nu}{\tau V_m^2}} - \frac{1}{2} \right).$$

Here, we find that

$$\frac{2(3\pi - 2\text{Arccos}(\sqrt{1 - \theta^2}))}{(1 + \theta)\sqrt{1 - \theta^2}} > 12.71$$

for $\theta \in (0, 1)$. Namely, we conclude that there exists $\tau V_m^2/\nu$ which satisfies (4.4) and $\tau V_m^2/\nu \leq 12.71$. For example, $\tau V_m^2/\nu = 4\pi$ satisfies these conditions.

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Declarations

Conflict of interest None.

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